

CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning

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Abstract

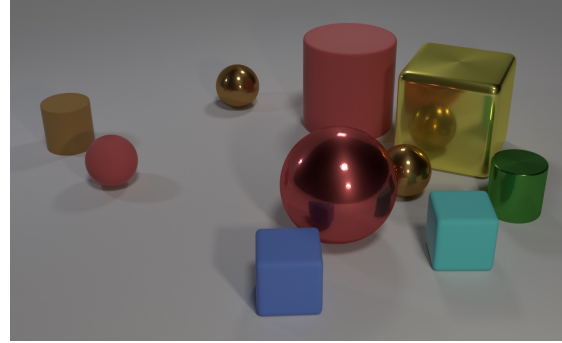
When building artificial intelligence systems that can reason and answer questions about visual data, we need diagnostic tests to analyze our progress and discover shortcomings. Existing benchmarks for visual question answering can help, but have strong biases that models can exploit to correctly answer questions without reasoning. They also conflate multiple sources of error, making it hard to pinpoint model weaknesses. We present a diagnostic dataset that tests a range of visual reasoning abilities. It contains minimal biases and has detailed annotations describing the kind of reasoning each question requires. We use this dataset to analyze a variety of modern visual reasoning systems, providing novel insights into their abilities and limitations.

1. Introduction

A long-standing goal of artificial intelligence research is to develop systems that can reason and answer questions about visual information. Recently, several datasets have been introduced to study this problem [4, 10, 21, 26, 32, 46, 49]. Each of these Visual Question Answering (VQA) datasets contains challenging natural language questions about images. Correctly answering these questions requires perceptual abilities such as recognizing objects, attributes, and spatial relationships as well as higher-level skills such as counting, performing logical inference, making comparisons, or leveraging commonsense world knowledge [31]. Numerous methods have attacked these problems [2, 3, 9, 24, 44], but many show only marginal improvements over strong baselines [4, 16, 48]. Unfortunately, our ability to understand the limitations of these methods is impeded by the inherent complexity of the VQA task. Are methods hampered by failures in recognition, poor reasoning, lack of commonsense knowledge, or something else?

The difficulty of understanding a system’s competences

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Q: Are there an equal number of large things and metal spheres?
Q: What size is the cylinder that is left of the brown metal thing that is left of the big sphere? Q: There is a sphere with the same size as the metal cube; is it made of the same material as the small red sphere?
Q: How many objects are either small cylinders or metal things?

Figure 1. A sample image and questions from CLEVR. Questions test aspects of visual reasoning such as attribute identification, counting, comparison, multiple attention, and logical operations.

is exemplified by *Clever Hans*, a 1900s era horse who appeared to be able to answer arithmetic questions. Careful observation revealed that Hans was correctly “answering” questions by reacting to cues read off his human observers [30]. Statistical learning systems, like those used for VQA, may develop similar “cheating” approaches to superficially “solve” tasks without learning the underlying reasoning processes [35, 36]. For instance, a statistical learner may correctly answer the question “What covers the ground?” not because it understands the scene but because biased datasets often ask questions about the ground when it is snow-covered [1, 47]. How can we determine whether a system is capable of sophisticated reasoning and not just exploiting biases of the world, similar to *Clever Hans*?

In this paper we propose a *diagnostic dataset* for studying the ability of VQA systems to perform visual reasoning. We refer to this dataset as the Compositional Language and Elementary Visual Reasoning diagnostics dataset (CLEVR; pronounced as *clever* in homage to Hans). CLEVR contains 100k rendered images and about one million automatically-generated questions, of which 853k are unique. It has chal-

lenging images and questions that test visual reasoning abilities such as counting, comparing, logical reasoning, and storing information in memory, as illustrated in Figure 1.

We designed CLEVR with the explicit goal of enabling detailed analysis of visual reasoning. Our images depict simple 3D shapes; this simplifies recognition and allows us to focus on *reasoning skills*. We ensure that the information in each image is *complete and exclusive* so that external information sources, such as commonsense knowledge, cannot increase the chance of correctly answering questions. We minimize question-conditional bias via rejection sampling within families of related questions, and avoid degenerate questions that are seemingly complex but contain simple shortcuts to the correct answer. Finally, we use structured ground-truth representations for both images and questions: images are annotated with ground-truth object positions and attributes, and questions are represented as *functional programs* that can be executed to answer the question (see Section 3). These representations facilitate in-depth analyses not possible with traditional VQA datasets.

These design choices also mean that while images in CLEVR may be visually simple, its questions are complex and require a range of reasoning skills. For instance, factorized representations may be required to generalize to unseen combinations of objects and attributes. Tasks such as counting or comparing may require short-term memory [15] or attending to specific objects [24, 44]. Questions that combine multiple subtasks in diverse ways may require compositional systems [2, 3] to answer.

We use CLEVR to analyze a suite of VQA models and discover weaknesses that are not widely known. For example, we find that current state-of-the-art VQA models struggle on tasks requiring *short term memory*, such as comparing the attributes of objects, or *compositional reasoning*, such as recognizing novel attribute combinations. These observations point to novel avenues for further research.

Finally, we stress that accuracy on CLEVR is not an end goal in itself: a hand-crafted system with explicit knowledge of the CLEVR universe might work well, but will not generalize to real-world settings. Therefore CLEVR should be used *in conjunction* with other VQA datasets in order to study the reasoning abilities of general VQA systems.

The CLEVR dataset, as well as code for generating new images and questions, will be made publicly available.

2. Related Work

In recent years, a range of benchmarks for visual understanding have been proposed, including datasets for image captioning [7, 8, 23, 45], referring to objects [19], relational graph prediction [21], and visual Turing tests [12, 27]. CLEVR, our diagnostic dataset, is most closely related to benchmarks for visual question answering [4, 10, 21, 26, 32, 37, 46, 49], as it involves answering natural-language

questions about images. The two main differences between CLEVR and other VQA datasets are that: (1) CLEVR minimizes biases of prior VQA datasets that can be used by learning systems to answer questions correctly without visual reasoning and (2) CLEVR’s synthetic nature and detailed annotations facilitate in-depth analyses of reasoning abilities that are impossible with existing datasets.

Prior work has attempted to mitigate biases in VQA datasets in simple cases such as yes/no questions [12, 47], but it is difficult to apply such bias-reduction approaches to more complex questions without a high-quality semantic representation of both questions and answers. In CLEVR, this semantic representation is provided by the functional program underlying each image-question pair, and biases are largely eliminated via sampling. Winograd schemas [22] are another approach for controlling bias in question answering: these questions are carefully designed to be ambiguous based on syntax alone and require commonsense knowledge. Unfortunately this approach does not scale gracefully: the first phase of the 2016 Winograd Schema Challenge consists of just 60 hand-designed questions. CLEVR is also related to the bAbI question answering tasks [38] in that it aims to diagnose a set of clearly defined competences of a system, but CLEVR focuses on visual reasoning whereas bAbI is purely textual.

We are also not the first to consider synthetic data for studying (visual) reasoning. SHRDLU performed simple, interactive visual reasoning with the goal of moving specific objects in the visual scene [40]; this study was one of the first to demonstrate the brittleness of manually programmed semantic understanding. The pioneering DAQUAR dataset [28] contains both synthetic and human-written questions, but they only generate 420 synthetic questions using eight text templates. VQA [4] contains 150,000 natural-language questions about abstract scenes [50], but these questions do not control for question-conditional bias and are not equipped with functional program representations. CLEVR is similar in spirit to the SHAPES dataset [3], but it is more complex and varied both in terms of visual content and question variety and complexity: SHAPES contains 15,616 total questions with just 244 unique questions while CLEVR contains nearly a million questions of which 853,554 are unique.

3. The CLEVR Diagnostic Dataset

CLEVR provides a dataset that requires complex reasoning to solve and that can be used to conduct rich diagnostics to better understand the visual reasoning capabilities of VQA systems. This requires tight control over the dataset, which we achieve by using synthetic images and automatically generated questions. The images have associated ground-truth object locations and attributes, and the questions have an associated machine-readable form. These

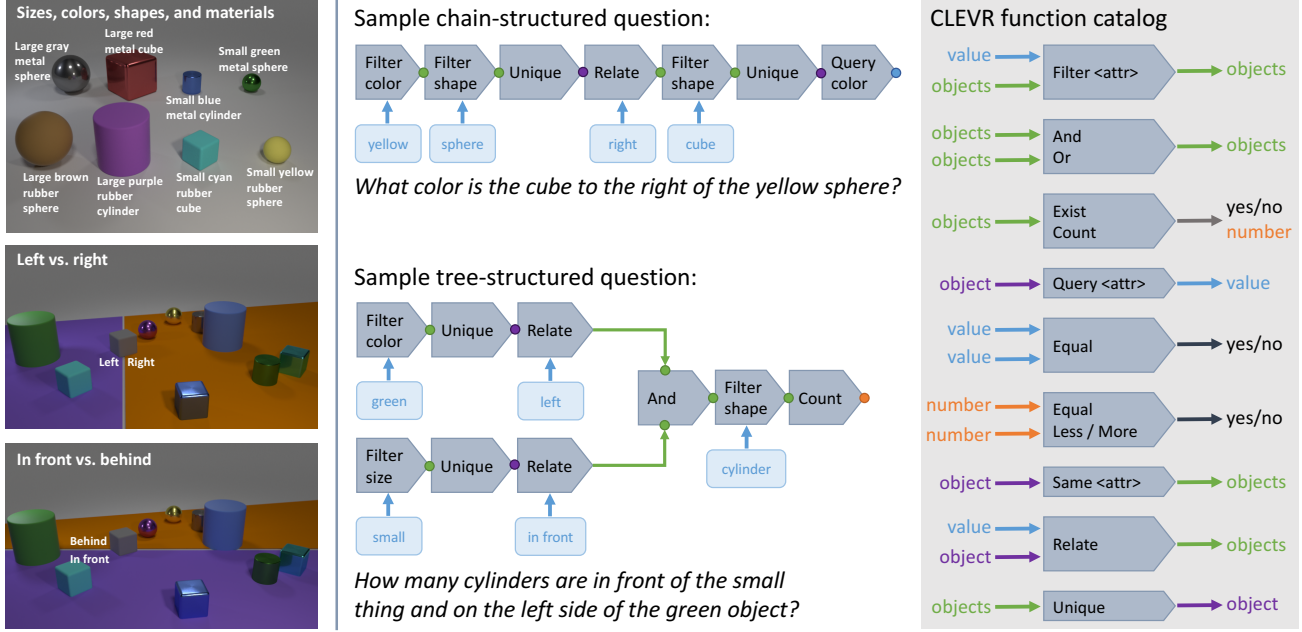


Figure 2. A field guide to the CLEVR universe. **Left:** Shapes, attributes, and spatial relationships. **Center:** Examples of questions and their associated functional programs. **Right:** Catalog of basic functions used to build questions. See Section 3 for details.

ground-truth structures allow us to analyze models based on, for example: question type, question topology (chain vs. tree), question length, and various forms of relationships between objects. Figure 2 gives a brief overview of the main components of CLEVR, which we describe in detail below.

Objects and relationships. The CLEVR universe contains three object shapes (cube, sphere, and cylinder) that come in two absolute sizes (small and large), two materials (shiny “metal” and matte “rubber”), and eight colors. Objects are spatially related via four relationships: “left”, “right”, “behind”, and “in front”. The semantics of these prepositions are complex and depend not only on relative object positions but also on camera viewpoint and context. We found that generating questions that invoke spatial relationships with semantic accord was difficult. Instead we rely on a simple and unambiguous definition: projecting the camera viewpoint vector onto the ground plane defines the “behind” vector, and one object is behind another if its ground-plane position is further along the “behind” vector. The other relationships are similarly defined. Figure 2 (left) illustrates the objects, attributes, and spatial relationships in CLEVR. The CLEVR universe also includes one non-spatial relationship type that we refer to as the *same-attribute relation*. Two objects are in this relationship if they have equal attribute values for a specified attribute.

Scene representation. Scenes are represented as collections of objects annotated with shape, size, color, material, and position on the ground-plane. A scene can also be represented by a *scene graph* [17, 21], where nodes are objects annotated with attributes and edges connect spatially related

objects. A scene graph contains all ground-truth information for an image and could be used to replace the vision component of a VQA system with *perfect sight*.

Image generation. CLEVR images are generated by randomly sampling a scene graph and rendering it using Blender [6]. Every scene contains between three and ten objects with random shapes, sizes, materials, colors, and positions. When placing objects we ensure that no objects intersect, that all objects are at least partially visible, and that there are small horizontal and vertical margins between the image-plane centers of each pair of objects; this helps reduce ambiguity in spatial relationships. In each image the positions of the lights and camera are randomly jittered.

Question representation. Each question in CLEVR is associated with a *functional program* that can be *executed* on an image’s scene graph, yielding the answer to the question. Functional programs are built from simple basic functions that correspond to elementary operations of visual reasoning such as *querying* object attributes, *counting* sets of objects, or *comparing* values. As shown in Figure 2, complex questions can be represented by compositions of these simple building blocks. Full details about each basic function can be found in the supplementary material.

As we will see in Section 4, representing questions as functional programs enables rich analysis that would be impossible with natural-language questions. A question’s functional program tells us exactly which reasoning abilities are required to solve it, allowing us to compare performance on questions requiring different types of reasoning.